Muhammad Abdullah Haroon

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Education

National University of Computer and Emerging Sciences (FAST-NUCES) — **Lahore, Pakistan**
Bachelor of Science in Computer Science (BSCS) — *Aug 2023 – Expected 2027*
FYP: Multi-Domain, Bi-Script Fake News Detection & Deepfake Audio Detection for Urdu

Work Experience

FAST School of Computing — **Jul 2026 – Sep 2026**
Research Intern — *Lahore, Pakistan*

- Led two independent research studies on low-resource **Urdu NLP** under the supervision of Dr. Maryam Bashir, formalized into full papers on cross-dataset generalization and domain robustness.
- Extended an independently-established Urdu FND finding (a length-confound shortcut driving **macro-F1 collapse from 0.771 to 0.005**) by adding English FND and sarcasm-detection domains under a unified zero-shot transfer protocol (**XLM-RoBERTa** vs. **TF-IDF + LR/SVM**), confirming the failure mode generalizes across languages and tasks.
- Built and curated a **33-dataset, 288,899-example** Urdu benchmark across sentiment, abusive language, fake news, and QA-validation tasks; ran a deduplication pass removing **4,196** train-test leaks prior to release on Hugging Face, Zenodo, and GitHub.
- Benchmarked **5 models** (**XLM-RoBERTa, mBERT, Llama-3.1-8B, Qwen-2.5-7B, Mistral-7B**) under a baseline-guarded relative-robustness metric, surfacing a 3-tier task hierarchy from **84.5–97.3%** (sentiment) down to **48.1–64.4%** (fake news / hate speech).

QuickFuel — **Jun 2026 – Jul 2026**
IT Intern — *Lahore, Pakistan*

- First person in company history – not just among interns – to introduce **AI/ML** work; spoke up and stood out in a traditionally .NET/WPF-focused engineering team to drive the addition of AI capabilities.
- Audited the company's Management Portal, identifying **9 AI integration opportunities** and prioritizing the top 3 by feasibility; ran an HCI heuristic evaluation surfacing **6 usability violations**, reported to and presented before senior leadership.
- Built a sentiment analysis prototype (**Python, LLMs, FastAPI, PostgreSQL**) and a semantic similarity search system on **ChromaDB** for context-aware product retrieval (e.g., resolving "bottle of water" to related beverages) in place of keyword matching.
- Built foundational production experience in **C#, XAML, .NET, WPF, MVVM**, and Dependency Injection while self-directing the AI workstream in parallel.

Optimus Automate — **Jun 2026 – Jul 2026**
AI Automation Intern — *Remote*

- Built an **LLM-powered** email triage and response agent integrated with the **Gmail API**: classified inbound mail by intent (inquiry, complaint, support) and auto-drafted context-aware replies.
- Developed a **RAG-based** document Q&A and summarization tool for PDFs/text files, grounding every answer in retrieved source content to avoid hallucinated responses on contracts and technical reports.
- Delivered both agents as production-style tools using an LLM + retrieval architecture aimed at removing repetitive manual triage and document-review work.

Research

Domain Robustness of Multilingual NLP Models Across Urdu and Roman Urdu Scripts

M. A. Haroon, M. Bashir | *In preparation – final stages; targeting Springer Language Resources and Evaluation*

- Released **33 Urdu/Roman Urdu** text-classification datasets (**288,899** labeled examples) across sentiment, abusive language, fake news, and QA-validation tasks, publicly released on Hugging Face, Zenodo (DOI: 10.5281/zenodo.22195610), and GitHub.
- Defined a domain-robustness evaluation protocol (same-source SS, target-domain TT, cross-domain ST, Source/Target Drop, baseline-guarded %Rob) and removed **4,196** exact-match train–test leaks before evaluation.
- Evaluated **5 models** (XLM-RoBERTa, mBERT, Llama-3.1-8B, Qwen-2.5-7B, Mistral-7B), establishing a 3-tier task hierarchy: **84.5–97.3%** relative robustness on sentiment vs. **48.1–64.4%** on fake news/hate speech.

When Benchmarks Mislead: Shortcut Learning, Length Confounds, and the Limits of Cross-Dataset Generalization in Multilingual Fake News and Sarcasm Detection

M. A. Haroon | arXiv:2607.14131

- Ran a unified zero-shot cross-dataset transfer study across **3 domains** (Urdu FND, English FND, sarcasm detection), fine-tuning **XLM-RoBERTa** against TF-IDF + LR/SVM baselines on each.
- Identified a length confound in the Ax-to-Grind Urdu dataset (fake articles average **3.4**× longer than real), driving a transfer collapse to **macro-F1 = 0.005** in one direction against **0.771** in reverse.
- Reproduced the same shortcut-learning failure on English FND (WELFake vs. ISOT) and sarcasm detection (TweetEval Irony vs. Sarcasm Corpus V2), and packaged the diagnostic method as a reusable protocol for any binary text-classification setting.

Bitcoin Price Direction Prediction via Regime-Aware Multi-Modal Fusion of Social Sentiment and Technical Features

M. A. Haroon | arXiv:2607.23370

- Proposed **Regime-Aware Multi-Modal Learning (RAML)**: a learnable sigmoid gate conditioning the fusion of FinBERT-scored Reddit sentiment and BiLSTM-encoded OHLCV features on a rolling 24-hour volatility regime.
- Evaluated on **3,491** hourly Bitcoin observations (Jul 2024 – Sep 2025) against 3 baselines; RAML reached **macro-F1 of 0.5474 (3h) / 0.5513 (6h)** and the highest 3h AUC (0.5084).
- Ran a component ablation showing static concatenation causes 6h recall to collapse (F1 drops to 0.14) – evidence regime-conditioned fusion is necessary, not incidental.

Projects

TaxChronos | Python, LangChain, Qdrant, Gemini API, RAGAS GitHub
Multilingual RAG system for Pakistani tax queries with English–Urdu code-switching

- Built a dense-retrieval RAG pipeline (**Qdrant** + **Gemini**) that answers tax queries in **under 20 seconds** versus 2–3 minutes of manual document search, measured on a held-out code-switched test set.
- Constructed a code-switched English–Urdu evaluation set and computed baseline **RAGAS** scores (Faithfulness, Answer Relevancy, Context Precision, Context Recall) against a naive-retrieval pipeline.

Daily Digest | FastAPI, PostgreSQL (Neon), Gemini, JWT, Docker, Resend GitHub
AI news aggregator delivering personalized, scheduled email digests – live in production

- Deployed a Generative AI product live in production with **15+ active users**, running an unattended fetch → rank → summarize → email pipeline entirely on free-tier infrastructure.
- Built a custom Pakistan-relevance ranking function combined with **Gemini**-generated summaries and per-article caching in Postgres to cut redundant LLM calls.
- Engineered a per-user scheduled delivery system (**APScheduler** + JWT auth) supporting 5 delivery windows and a keep-alive layer to defeat free-tier cold starts.

Urdu Domain-Robustness Benchmark | Python, HF Transformers, XLM-RoBERTa GitHub

Open-source benchmark repository and evaluation tooling backing the Domain Robustness research above

- Engineered the full data pipeline behind the **33-dataset, 8-task**, dual-script benchmark, including deduplication, SS/ST split construction, and scoring code released alongside the paper.
- Open-sourced **33 datasets** and **5** fine-tuned/evaluated NLP models across Hugging Face, Zenodo, and GitHub, with reusable evaluation harnesses for benchmarking new models under the same protocol.

Bilingual WhisperBench | Python, OpenAI Whisper, FastAPI, SQLAlchemy GitHub
Zero-shot benchmarking dashboard for compact Whisper models on code-mixed speech

- Ran a zero-shot evaluation of **Whisper Tiny/Base/Small** (39M–244M params) on **327** curated Urdu-English code-mixed recordings from **11** bilingual speakers, stratified across 3 code-mixing intensity tiers.
- Introduced the **Switch Point Error Rate (SPER)** metric to isolate transcription failures at language-switch boundaries, and identified Whisper Small + English-forced inference as the best zero-shot configuration (**WER 0.899**, CV 10.72%).

FinSight | React.js, FastAPI, Scikit-learn, PostgreSQL GitHub
Financial behavior and risk analysis platform

- Designed a weighted multi-factor risk score (volatility 30%, anomalies 25%, trend 20%, patterns 15%, amounts 10%) fusing an **Isolation Forest** anomaly detector, an **HMM** behavioral-state pipeline, and an NLP transaction analyzer.
- Built the end-to-end CSV-to-dashboard pipeline classifying uploaded transaction data into **Low** / **Medium** / **High** / **Extreme** risk bands, with quarterly trend analytics and PDF/CSV/Excel export.

Technical Skills

Languages: Python, C, C++, C#, SQL, JavaScript, TypeScript, HTML, CSS
Machine Learning & Generative AI: PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, LangChain, RAG, Ollama, Prompt Engineering, Whisper, LSTM, n8n
Databases: PostgreSQL, MySQL, MS SQL Server, SQLite, MongoDB, Qdrant, ChromaDB, FAISS
Backend & APIs: FastAPI, SQLAlchemy, REST APIs, Node.js, Express.js, .NET
Frontend & Tooling: React.js, Tailwind CSS, Git/GitHub, Docker, Render, Jupyter/Colab, Antigravity, Cursor, VS Code

Certifications

AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents – Udemy
(view certification)
Machine Learning with Python – IBM
(view certification)
Data Analysis and Visualization with Python – Microsoft
(view certification)
Introduction to Generative AI – Google
(view certification)

Leadership & Extracurricular Activities

Team Leadership & Scrum Master – **FYP** and **semester projects** teams: ran sprint planning, delivered project presentations, and took ownership of distributing and managing tasks across multi-member teams.
Competitions, Teaching & Campus Involvement – **SQL Query Master Competition** SOFTEC'25 and **Media Officer** for SOFTEC; **ACM Speed Programming Competition**; delivered **AI masterclasses/workshops** on the foundations of AI; taught **MS Office** at Pact College (local institution); **Deputy Head of Public Relations**, FAST Business Club.